

The Dependence of Presentation Slides and Its Affect in Praxis Lectures

In February 2003, the Space Shuttle *Columbia* disintegrated in the sky. The spaceship was launched in January, and during lift-off, foam insulation broke off from an external tank and created a hole in the left wing of the shuttle. Upon re-entering the atmosphere, the hole led to an exchange of temperature between the internal shuttle and external gases, destroying the wing, the shuttle, and killing all 7 crew members as the shuttle broke apart [1]. NASA then put all their launches on a two-and-a-half-year hiatus while investigating the incident.

3 reports in the form of 28 Microsoft PowerPoint slides were presented to NASA by the Boeing engineers and due to poor structuring of data presentation, NASA management was falsely led to believe that the ship was okay and continued its operation, leading to the fatal re-entry [2]. Through Professor Chong, the class viewed the analysis of a PowerPoint slide by Edward Tufte, a professor at Yale University and an author of books about visual representation.

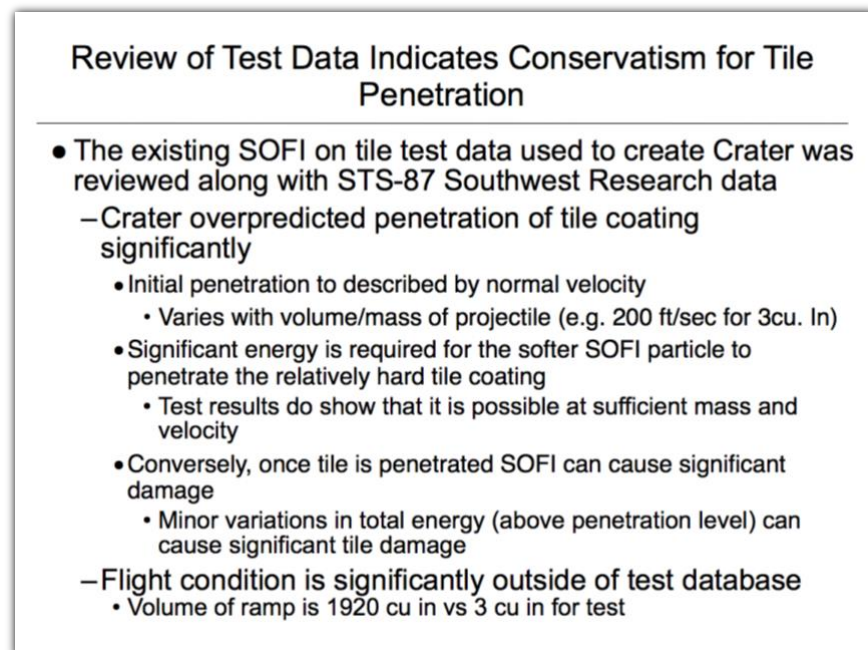


Figure 1: A slide from the reports of Boeing and presented by Professor Chong in class. Following definitions: SOFI – Spray on Foam Insulator; Crater – computer program used to simulate the damage on the wing; Ramp – foam debris that hit the wing. [3]

The analysis from Tufte covered issues such as text size, misuse of bullet points, inconsistent notation, and misleading vocabulary, all leading NASA to believe that there were no major concerns [2]. In class, I was led the believe that the slide shown by Professor Chong was used for a presentation from Boeing summarizing their analysis to NASA officials. Upon further research, I was shocked to find that the PowerPoint presentation was the report itself and that professional engineers had chosen PowerPoint to report such a serious matter.

PowerPoint’s automatic formatting of bullet points and sub-bullet points with smaller font sizes unintentionally created a hierarchy that made sentences with larger text seem more important and memorable. In the case of the Boeing presentations, smaller bullets that acknowledged doubts and uncertainties were seen as footnotes for the big bullets that did not address the issues of the damages [2], leading NASA to believe that the ship was good to continue the mission.

Analyzing PowerPoint (or Keynote) presentations of a first year Engineering Science physics class, Professor Thywissen effectively uses slides to enhance the knowledge of the course. **Figures 2 and 3** follows “visual reasoning” as explained by Tufte, where information is taken in more effectively when “relevant evidence is shown *adjacent in space* within out eye span” [2]. It is not filled with text as the slides are merely an add to the concepts taught in class.

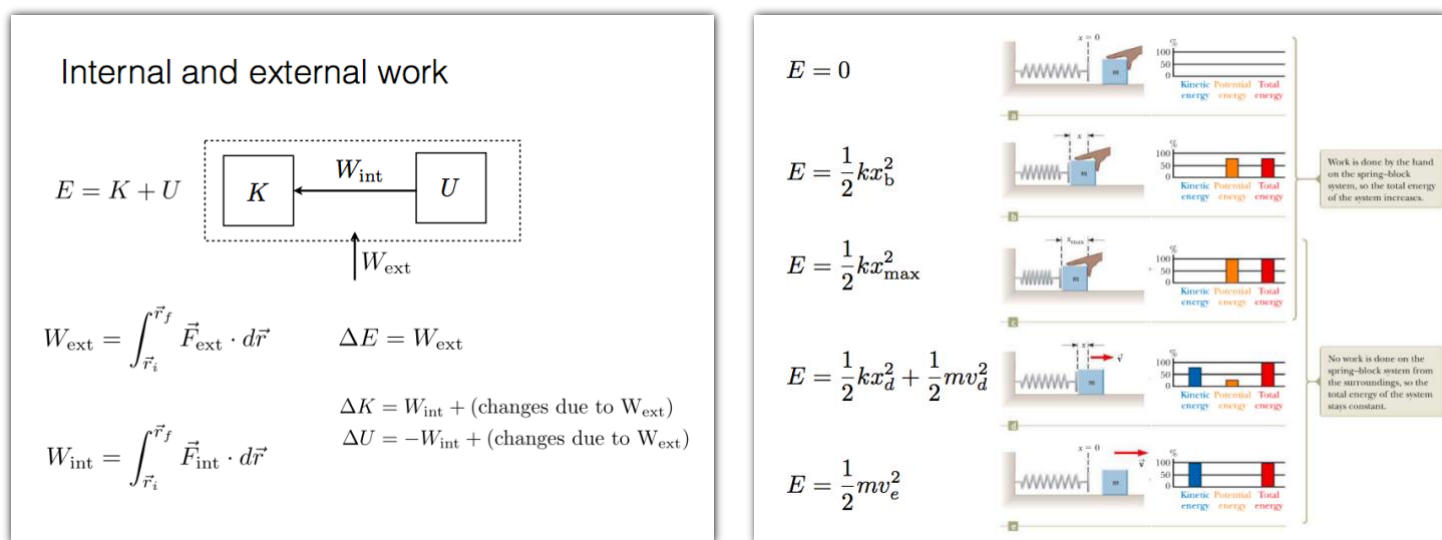


Figure 2 and 3: Thywissen’s slides are not text heavy and have diagrams to aid the predictive models [4].

Praxis lectures slides are what I call “non-linear”, meaning reading slides from top to bottom is not necessarily the order the information appears in, as in **Figure 4 and 5**. With the fast pace of lectures, students are advised to go through the lectures in their own time, as well as using them to prepare for upcoming assignments, the midterm, and the exam. However, due to the non-linear appearance of text bubbles, as well as the use of overlapping bubbles on top of text, uploaded slides are often over 40 pages long. The issue, as explained by Tufte, is that audiences “endure a relentless sequentially” and that that the surge of information in such a short time makes it difficult to take it all in and understand everything [2]. Slides such as **Figure 6** is an example of an overwhelming text that is not given enough time to take in during lecture.

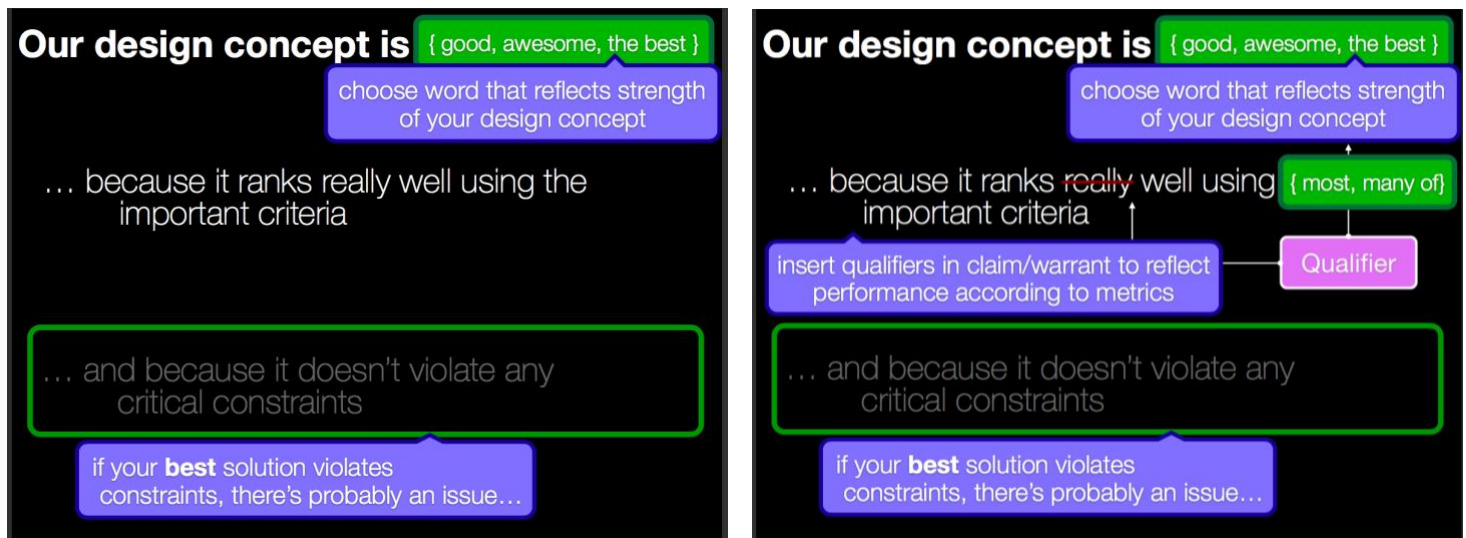


Figure 4 and 5: Successive slides of Lecture 29. The comments in text bubbles do not appear in order of appearance from top to bottom. [5]

We are **not** saying “do not attempt to simulate disability”.
We are saying that you need to be **deliberate in choosing**
how to model your stakeholders (including the disabled), in how
you **validate** your modelling, and in how you **frame** your work.

You did not experience having osteoarthritis.
You did not experience the effects of osteoarthritis on grip.
You did experience a treatment intended to simulate the effects of
osteoarthritis on grip strength **for the purpose of completing a
preliminary assessment of a design intended to improve the
writing experience of people with osteoarthritis.**

**In Praxis you must (RFC2119) with records
test your { design, metric, treatment } on
your team before testing on other proxies or
stakeholders**

Keywords: ableism, disability awareness, disability simulation, perspective taking

Figure 6: Text heavy slide with varying sizes and typefaces make it hard to read at a first glance and the speed of lectures does not allow me to absorb all the information [5].

Tufte recommends handouts aided by PowerPoints with images and videos that cannot be printed. Handouts should be given to the audience so they can go at their own pace, more effectively show text, numbers, etc., and provide a permanent record for view. The issue with Praxis slides is that they are both presentation slides meant for a lecture as well as handouts. This affects lectures as some slides may be flooded with text, and makes them inefficient as handouts as some slides require a speaker to guide the reader through non-linear slides.

In conclusion, like the Boeing analysis of the Columbia Spaceship, the Praxis lecture slides are not the most effective tool to convey the information needed. A solution could do what Professor Thywissen

does, uploading notes and slides separately. This is an issue that I, as well as my peers, have with Praxis lectures, and I believe addressing this will benefit future Engineering Science students in their learning of Praxis and engineering communication.

Appendix A: Source Extracts

[1] S. Dowling, "What Caused the Space Shuttle Columbia Disaster?", *BBC*, 2015.

“Their fate had been sealed 16 days before, when the Shuttle lifted off from Cape Canaveral. A piece of foam insulation the size of a briefcase fell off the spacecraft’s external fuel tank, cracking some of the heat-resistant tiles on the Shuttle’s left wing.”

“But what was recovered was enough to piece together the Shuttle’s final minutes. As the spacecraft descended to Earth, travelling thousands of miles an hour, super-heated atmospheric gases rushed through the breach in the heat-resistant tiles. These tiles were designed to withstand temperatures as high as 1,650C (3,000F) as the spacecraft re-entered the Earth’s atmosphere. Super-heated atmospheric gases forced their way into the Shuttle’s airframe through the hole in the wing, causing the Shuttle to destabilise and then disintegrate.”

[2] E. Tufte, *The Cognitive Style of Powerpoint: Pitching Out Corrupts Within*, 2nd ed. Cheshire: Graphics Press LLC, 2006.

“To help NASA officials assess the threat, Boeing Corporation engineers quickly prepared 3 reports, a total of 28 PowerPoint slides, dealing with the debris impact. These reports provided mixed readings of the threat to the Columbia; the lower-level bullets often mentioned doubts and uncertainties, but the highlighted executive summaries and big-bullet conclusions were quite optimistic. Convinced that the reports indicated no problem rather than uncertain knowledge, high-level NASA officials decided that the Columbia was safe and, furthermore, that no additional investigations were necessary...”

“In practice, PP slides are very low resolutions compared to paper, many computer screens, and the immense visual capacities of the human eye-brain system. With little information per slide, many many slides are needed. Audiences endure a relentless sequentially, one damn slide after another. Information stacked in time makes it difficult to understand context and evaluate relationships. Visual reasoning usually works more effectively when the relevant evidence is shown *adjacent in space* within out eye span. This is especially the case for statically data, where the fundamental analytical task is to make comparisons.”

“High-resolution handouts allow viewers to contextualize, compare, narrate, and recast evidence. In contrast, data-thin, forgetful displays tend to make audiences ignorant and passive, and also to diminish the credibility of the presenter...

It is helpful to provide audience members with at least one mode of information that allows them to control the order and pace of learning— unlike slides and unlike talk. Paper handouts for talks will help provide a permanent record for review—again unlike projected images and talk...

Paper handouts at a talk can effectively show text, numbers, data graphics, images. Printed materials, which should largely replace PP, bring information transfer rates in presentations up to that of everyday material in newspapers, magazines, books, and internet screens.”

[3] P. Parker, D. Chao, I. Norman and M. Dunham, "Orbiter Assessment of STS-107 ET Bipod Insulation Ramp Impact", NASA, 2003.

Review of Test Data Indicates Conservatism for Tile Penetration

- The existing SOFI on tile test data used to create Crater was reviewed along with STS-87 Southwest Research data
 - Crater overpredicted penetration of tile coating significantly
 - Initial penetration to described by normal velocity
 - Varies with volume/mass of projectile (e.g. 200 ft/sec for 3cu. In)
 - Significant energy is required for the softer SOFI particle to penetrate the relatively hard tile coating
 - Test results do show that it is possible at sufficient mass and velocity
 - Conversely, once tile is penetrated SOFI can cause significant damage
 - Minor variations in total energy (above penetration level) can cause significant tile damage
 - Flight condition is significantly outside of test database
 - Volume of ramp is 1920 cu in vs 3 cu in for test

[4] J. Thywissen, "Internal Forces & Potential Energy", University of Toronto, 2017.

Internal and external work

$E = K + U$

K

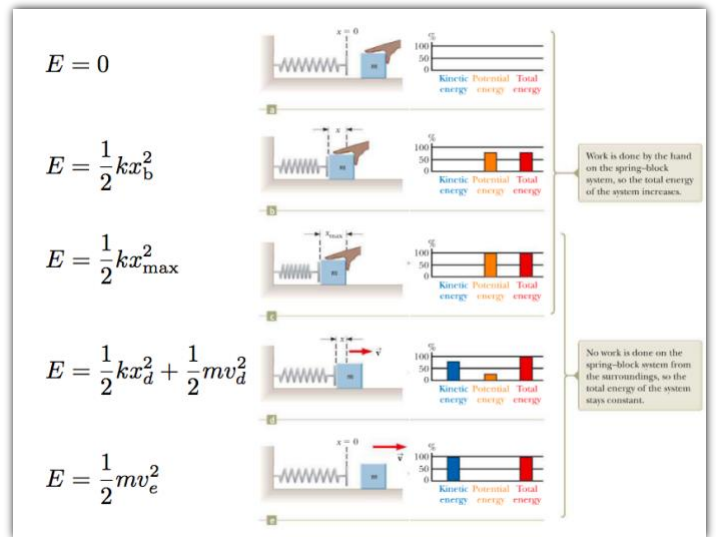
$\xleftarrow{W_{\text{int}}}$

U

$\uparrow W_{\text{ext}}$

$$W_{\text{ext}} = \int_{\vec{r}_i}^{\vec{r}_f} \vec{F}_{\text{ext}} \cdot d\vec{r} \quad \Delta E = W_{\text{ext}}$$

$$W_{\text{int}} = \int_{\vec{r}_i}^{\vec{r}_f} \vec{F}_{\text{int}} \cdot d\vec{r} \quad \Delta K = W_{\text{int}} + (\text{changes due to } W_{\text{ext}})$$

$$\Delta U = -W_{\text{int}} + (\text{changes due to } W_{\text{ext}})$$


[5] J. Foster and A. Chong, "ESC101 20179 Lecture 29", University of Toronto, 2017.

Our design concept is { good, awesome, the best }

choose word that reflects strength of your design concept

... because it ranks really well using the important criteria

... and because it doesn't violate any critical constraints

if your **best** solution violates constraints, there's probably an issue...

Our design concept is { good, awesome, the best }

choose word that reflects strength of your design concept

... because it ranks **really** well using { most, many of } important criteria

insert qualifiers in claim/warrant to reflect performance according to metrics

Qualifier

... and because it doesn't violate any critical constraints

if your **best** solution violates constraints, there's probably an issue...

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